Derivation of the bound on the generalization cost

$$P\left\{\sup_{\underline{w}\in\mathcal{L}} \left|R[\underline{w}] - R^{(P)}_{emp}[\underline{w}]\right| > \eta\right\} < \underbrace{4\exp\left\{G^{\mathcal{L}}_{(2_P)} - P\left(\eta - \frac{1}{P}\right)^2\right\}}_{\varepsilon}$$

With probability ε:

There is at least one model $\underline{w}$, for which

$$\left|R[\underline{w}] - R^{(P)}_{emp}[\underline{w}]\right| > \eta$$

With probability $1-\varepsilon$

$$\left|R[\underline{w}] - R^{(P)}_{emp}[\underline{w}]\right| \leq \eta \quad \text{for all models } \underline{w}$$

Let's assume: $R[\underline{w}] \geq R^{(P)}_{emp}[\underline{w}]$

$$\Rightarrow \quad R[\underline{w}] \leq R^{(P)}_{emp}[\underline{w}] + \eta$$

Let's assume: $R[\underline{w}] < R^{(P)}_{emp}[\underline{w}]$ ← non-negative

$$\Rightarrow \quad R[\underline{w}] \leq R^{(P)}_{emp}[\underline{w}] + \eta$$

Therefore it holds:

$$\Rightarrow \quad R[\underline{w}] \leq R^{(P)}_{emp}[\underline{w}] + \eta$$

for all models $\underline{w}$ with probability larger/equal $1-\varepsilon$

Calculation of η in terms of ε:

$$4\exp\left(G^{\mathcal{L}}_{(2_P)} - P\left(\eta - \frac{1}{P}\right)^2\right) = \varepsilon$$

$$G^{\mathcal{L}}_{(2_P)} - P\left(\eta - \frac{1}{P}\right)^2 = \ln\frac{\varepsilon}{4}$$

$$-P\left(\eta - \frac{1}{P}\right)^2 = \ln\frac{\varepsilon}{4} - G^{\mathcal{L}}_{(2_P)}$$

$$\left(\eta - \frac{1}{P}\right)^2 = \frac{1}{P}\left(G^{\mathcal{L}}_{(2_P)} - \ln\frac{\varepsilon}{4}\right)$$

$$\eta = \sqrt{\frac{1}{P}G^{\mathcal{L}}_{(2_P)} - \ln\frac{\varepsilon}{4}} + \frac{1}{P}$$